Highlights

Operating a gas-lifted well under severe slugging: time-averaged signatures from a two-fluid simulator, physics-informed surrogates of the choke and gas-lift decisions, and their operational optimisation

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- Gas-lift branch simulated with the two-fluid model, controlled by the platform gas pressure
- Warm-started Sobol' campaign with adaptive horizon: 156 runs over choke and lift gas in a day
- Time-averaged markers at eight points replace trajectories as the training data
- Physics of the averages lifts the surrogate of the mean liquid rate from R^2 0.71 to 0.89
- Stability boundary from 80 to 125 bar of lift gas; spans of the rates remain hard to predict

Operating a gas-lifted well under severe slugging: time-averaged signatures from a two-fluid simulator, physics-informed surrogates of the choke and gas-lift decisions, and their operational optimisation

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ABSTRACT

A gas-lifted satellite well with a flowline and a riser can produce in severe slugging, in a pressure cycle or in a steady state depending on two operating decisions, the opening of the topside choke and the pressure of the lift gas at the platform, and the time-averaged production of an unstable operating point is not what a steady-state model predicts for it. This paper builds the data and the surrogate models of those two decisions and uses them to optimise the operation of the well. The well, flowline, riser and gas-lift branch are simulated with the four-equation two-fluid model of a companion study, with the flow pattern identified dynamically and the gas-lift line, annulus and check valve simulated with the same model and controlled only by the platform gas pressure. A sampling campaign over the two decisions, in rounds of Sobol' points run as warm-started chains along the shortest path from the base case, with an adaptive horizon that detects the established regime and captures three cycles, produced 156 runs in a day on a laptop; each run is reduced to time-averaged markers (mean, minimum, maximum and the two extreme frequencies) of pressure, void fraction and phase flow rates at eight characteristic points of the network. The map of the decisions shows a stability boundary that runs from 80 bar of lift gas with the choke nearly closed to 125 bar with it open, continuous injection modulated by the cycle below it, and a production gain of up to 34 % above it. Two surrogates of the mean liquid, gas and lift-gas rates and of their spans were trained on the markers: a neural network and a physics-informed network whose physics are the time-averaged mass balances of the well, the gas-lift branch and the valve, exact for the averages and evaluated at collocation points without simulation. The physics raise the test coefficient of determination of the mean liquid rate from 0.71 to 0.89 and those of the gas and lift-gas rates to 0.997, halve the error with ten to forty training runs, and leave the spans, the signature of slugging, as the quantity that neither network predicts well. The surrogate and the stability map then enter an operational optimisation of the two decisions — revenue of the mean liquid rate minus the cost of compression, under separator and pressure-swing constraints — whose results are pending.

1. Introduction

Continuous gas lift is the artificial-lift method of most deepwater satellite wells in Brazil, and its optimisation — how much gas to inject in each well of a field that shares a compressor — has a large literature built on steady-state performance curves (Souza, 2010). A well with a flowline and a riser, however, does not always have a steady state to be optimised: with the choke open and little gas it produces in severe slugging (Schmidt, Brill and Beggs, 1980; Taitel, 1986; Jansen, Shoham and Taitel, 1996), and the time-averaged production, the liquid surges that reach the separator and the pressure swings at the wellhead are quantities of the cycle, not of the unstable steady state that a nodal model would return for the same point. A companion study (Souza and Costa, 2026) developed a four-equation two-fluid model with dynamic flow-pattern identification that reproduces the cycle in a well–flowline–riser system of field dimensions, located the onset of slugging as the choke opens and showed, with a gas-lift branch added to the system, that injection at 100 to 120 bar regularises the cycle and from 130 bar suppresses it.

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This paper is about the operation of that gas-lifted well. Its question is practical: given the two decisions an operator has, the opening of the topside choke and the pressure of the lift gas at the platform, what does the well deliver on average, with what surges and in which regime — and can a model cheap enough for optimisation answer that over the whole plane of the two decisions? The dynamic simulation is too expensive to be called inside an optimiser (minutes per point), so the paper builds a surrogate, and it does so with a method that follows the companion work on compositional steady-state surrogates (Carvalho, Costa, Nunes and Souza, 2026): a multilayer perceptron trained on simulated data, and the same network with a physics term (Raissi, Perdikaris and Karniadakis, 2019; Karniadakis, Kevrekidis, Lu, Perdikaris, Wang and Yang, 2021). The novelty is on both sides of the surrogate. On the data side, the targets are not profiles but time-averaged signatures of a dynamic simulation, and producing them in affordable time required a campaign design of its own: an adaptive horizon that recognises the established regime, warm starts along a shortest path through the sampled points, and a storage that keeps only what the training needs. On the physics side, the governing equations cannot be imposed on a time average of a cycling system; what can, exactly, are the time-averaged mass balances of the well, of the gas-lift branch and of the valve, and the paper shows that this much physics is what the data alone cannot supply.

Section 2 summarises the phenomenological model and the gas-lift branch. Section 3 describes the campaign, the horizon, the markers and the resulting map of the decisions. Section 4 presents the two surrogates and Section 5 compares them. Section 6 formulates the operational optimisation on the surrogate and will report its results.

2. The phenomenological model

The system is the well–flowline–riser of the companion study (Souza and Costa, 2026) with its gas-lift branch (Fig. 1): a vertical well of 1500 m (127 mm) fed by a linear inflow performance relation, a flowline of 2000 m (152.4 mm) whose last 500 m descend at 2° to a low point, a bend of 5 m radius, a riser of 317.5 m and 10 m of horizontal pipe to a choke that discharges into a separator at 10 bar. The gas-lift branch is a 4-in line that runs from the platform along the route of the production line to the wellhead, the casing–tubing annulus (9 5/8 in by 5 1/2 in) to a gas-lift valve 30 m above the bottom of the well, and a check valve fed by a chamber at the base of the annulus. The base well has a reservoir pressure of 200 bar, a productivity index of 5 $\mathrm{m^3\,d^{-1}\,bar^{-1}}$ and a gas–oil ratio of 50 $\mathrm{m^3\,m^{-3}}$; the fluid is the light oil of Carvalho (2022) frozen as a polytropic pair.

Every pipe, the gas pipes included, is described by the one-dimensional four-equation two-fluid model of the companion study: mass and momentum of each phase, closures by flow pattern (stratified, intermittent, bubbly, annular) selected dynamically on the local state through smooth logistic switches, a low-void rule, an objective virtual-mass term and an interfacial pressure in the slug unit that keep the model hyperbolic along the simulated trajectories. The staggered finite-volume discretisation with phase-wise donor cells and the conservative variable-step implicit integration of that study are kept unchanged, with a compiled right-hand side and exact Jacobians; the base mesh has 15 cells in the well, 35 in the flowline–riser pipe, 19 in the gas-lift line and 10 in the annulus, 323 states in all. The gas-lift branch is controlled by one number, the gas pressure P_{gas} at the platform end of the line: the injected rate is not prescribed but follows from P_{gas}, the static gas columns of the line and the annulus, the valve law $q = 0.0075\,C_V\sqrt{\rho_G\,\Delta P}$ ($C_V = 1.5$, ΔP in bar, regularised at the origin) and the pressure of the well at the injection point, so that it responds to whatever the well does; at 20 bar the valve never opens and the system is the base case of the companion study with the branch inert. The reservoir gas is the gas of the well; the lift gas adds to it at the injection point.

3. Data: campaign, horizon and markers

3.1. Sampling the two decisions

The inputs are the choke opening $a \in [0.1, 1]$ and the platform gas pressure $P_{gas} \in [20, 160]$ bar, with the well fixed at its base state. Points are drawn from a scrambled Sobol' sequence (Sobol', 1967; Owen, 1998) in rounds of 25, so that the rounds together remain a single space-filling design, and each round is run as one chain: the first point is the one closest to the base case (choke open, 20 bar) and the run starts from the final state of the base run of the companion study (8000 s of severe slugging); every later point starts from the final state of the previous one, in the order of the shortest open path through the 25 points (nearest neighbour improved by 2-opt, on the inputs rescaled to $[-1, 1]$). The warm start spares the transient of a blocked start — three to five cycles in the companion study — and keeps the steps between consecutive runs small; a large drop of P_{gas} between neighbours is the one expensive transition, since the gas line and the annulus then drain towards the platform, and the shortest path avoids most of them. Each point is identified

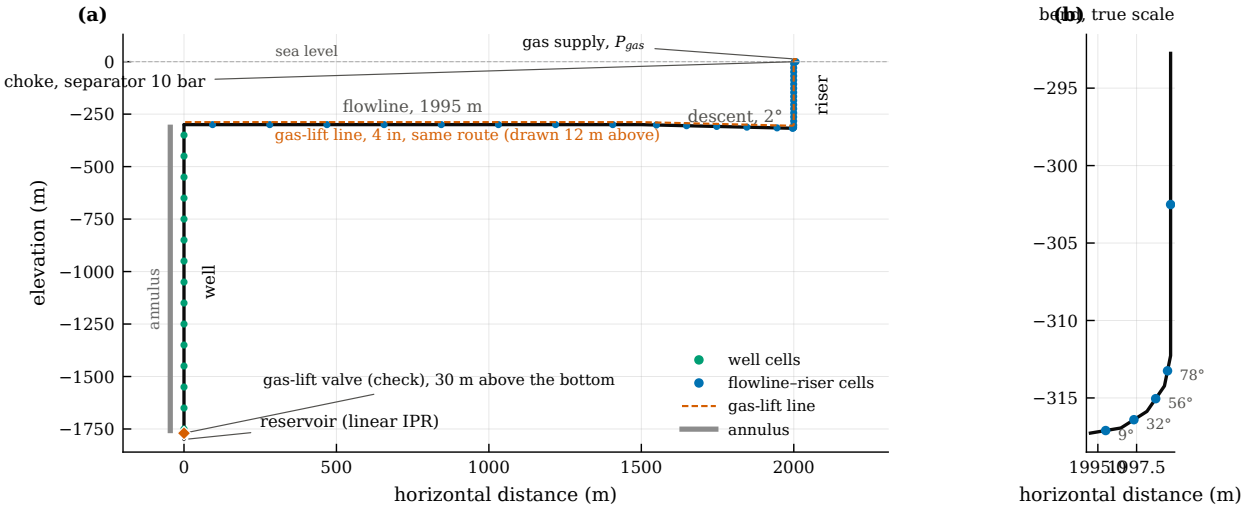


Figure 1: The well–flowline–riser system with the gas-lift branch (schematic, from the companion study): the gas-lift line follows the route of the production line, the annulus runs beside the well and the check valve injects 30 m above the bottom. The two operating decisions are the choke opening a and the platform gas pressure P_{gas}.

by a hash of its input tuple, and only what the training needs is stored: the markers below and the final state that seeds the next run.

3.2. Adaptive horizon with capture of cycles

Each run integrates in windows of 3000 s and monitors two signals, the pressure at the wellhead and the liquid rate through the choke. After a minimum time (4000 s from a blocked start, 1500 s from a warm start) the regime is taken as established when both signals are permanent (amplitude below 0.1 bar and 0.05 kg s^{-1}) or when two consecutive windows have the same envelope, that is, the same amplitude, mean, minimum and maximum within 10 % of the amplitude; the envelope criterion, rather than equality of periods, is what accepts the cycles whose blowouts alternate in size. The run then continues for three more cycles of the longest monitored period (600 s for a steady state), which are the capture window of the markers, and stops; a run that never settles stops at 24 000 s and is kept with the markers of its last 4000 s and a flag. A wall-clock budget of 40 min per run bounds the few points where the two horizontal cells between the riser and the choke switch flow pattern at every step, the one open point of the closures that the companion study reported; those runs are flagged and excluded from the training.

3.3. Markers at characteristic points

A run is reduced to markers of its capture window at eight points of the network: the bottom of the well, the injection point, the wellhead, the low point of the flowline, the top of the riser, the two ends of the gas-lift line and the base of the annulus. At each point four signals are recorded — pressure, void fraction, liquid and gas mass flow rates — and each signal is summarised by its mean, minimum, maximum and two frequencies, the inverse of the longest and of the shortest interval between successive maxima of the smoothed signal: equal for a periodic cycle, different when blowouts alternate, zero for a permanent signal. The gas injected through the valve is recorded in the same way, and a class is assigned from the wellhead pressure and the choke liquid rate as in the companion study (steady, pressure cycle or severe slugging when the peak liquid rate exceeds twice its mean). Trajectories are not kept: they would cost 12 MB per run against 10 kB of markers, and the surrogate consumes only the markers.

3.4. The map of the decisions

Table 1 summarises the campaign and Fig. 2 shows it on the plane of the decisions. Of the 156 runs, 83 produce in severe slugging, 10 in a pressure cycle and 63 in a steady state; 88 settled on a repeated cycle, 60 on a steady state, 2 reached the time limit and 6 the budget. The stability boundary is a diagonal that runs from about 80 bar of lift gas with the choke at 0.1 to about 125 bar with it open: the more open the choke, the more gas is needed to suppress the cycle. The check valve opens between 86 and 95 bar almost independently of the choke, so that below 90 bar the data

Table 1
The sampling campaign over the two decisions.

points (runs)	156
steady / pressure cycle / severe slugging	63 / 10 / 83
stop: steady / cycle captured / time limit / budget	60 / 88 / 2 / 6
simulated time per run (s): median, min–max	9640, 6600–24000
accepted steps per run: median, min–max	6546, 159–76960
integration time per run (min): median, max	4.2, 40.0

Base well of Part I ($P_{res} = 200\,\mathrm{bar}$, $J = 5\,\mathrm{m^3\,d^{-1}\,bar^{-1}}$, GOR 50) with the gas-lift branch; inputs sampled by scrambled Sobol' sequences in rounds of 25 over $P_{gas} \in [20, 160]$ bar and $a \in [0.1, 1]$; each round starts from the final state of the base run at the point closest to it and continues along the shortest path with warm starts. Budget: 40 min of integration per run.

are the base well with the choke as the only decision; between the opening of the valve and the boundary the injection is continuous and modulated by the cycle, which shortens and regularises; above the boundary the well is steady and its production rises with the gas, to $5.3\,\mathrm{kg\,s^{-1}}$ at 160 bar with the choke open, 34 % above the base case. The peak liquid rate through the choke reaches $45\,\mathrm{kg\,s^{-1}}$ in the slugging region with the choke open and falls below $5\,\mathrm{kg\,s^{-1}}$ in the steady region; the six runs that exhausted the budget sit on the boundary at high opening, where the riser receives the most gas.

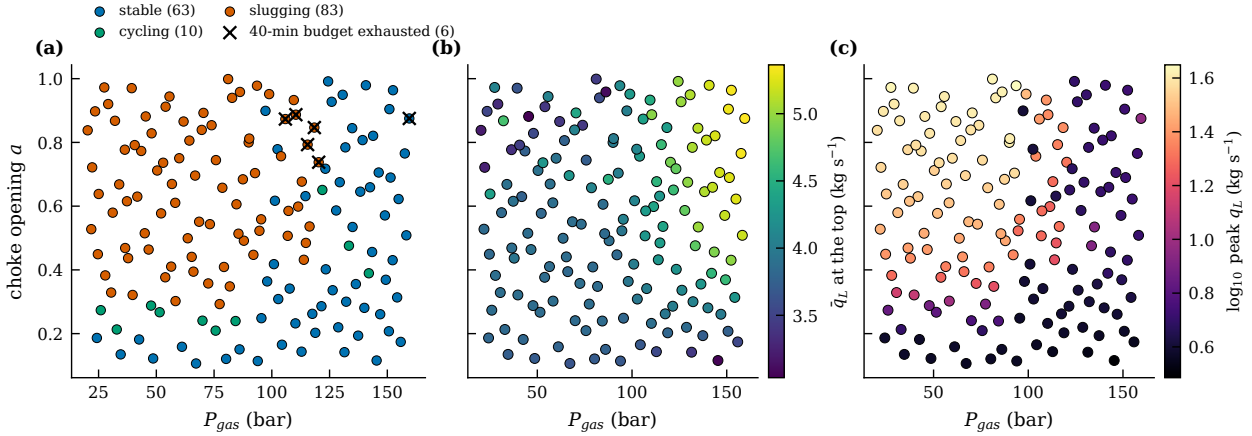


Figure 2: The 156 runs on the plane of the choke opening and the platform gas pressure: (a) class of the established regime, with the six runs that exhausted the 40-min integration budget before settling (crosses; their markers come from a partial trajectory and they are excluded from the training); (b) mean liquid rate through the choke; (c) peak liquid rate through the choke on a logarithmic scale.

4. Surrogates

4.1. Outputs and network

The surrogate maps the two decisions to nine time-averaged quantities: the mean liquid and gas rates at the top of the riser, the mean gas-lift rate through the valve, the spans (maximum minus minimum over the capture window) of the same three rates, which are the signature of slugging, and three auxiliary mean pressures — at the bottom of the well, at the injection point and at the base of the annulus — that the physics of the next section needs. The spans are learnt as log(1 + span), so that they are never negative and their orders of magnitude are resolved. The network is a multilayer perceptron (Hornik, Stinchcombe and White, 1989) with two hidden layers of 32 hyperbolic-tangent units and a linear output, inputs and outputs standardised with the training statistics, trained with Adam (Kingma and Ba, 2015) in full batch for 4000 epochs on the mean squared error of the standardised outputs, with the epochs unrolled in a compiled scan so that a network trains in about a second; 70 % of the 150 retained runs train it and 30 % test it. A

grid of one or two layers, 8 to 64 units and three learning rates (24 networks of each kind, Section 5) checked that the architecture is not critical.

4.2. Physics of the averages

A physics-informed network adds to the data loss the residual of the governing equations at collocation points that need not be simulated (Raissi et al., 2019). Here the targets are time averages of a dynamic system, and the time average of a cycle does not satisfy the two-fluid equations, which are nonlinear; what it satisfies exactly are the balances that are linear in the averaged quantities. Over a periodic cycle or a steady state, the mass of each phase in the system does not change on average, so that

$$R_1 = \bar{q}_L - J_L\,(P_{res} - \bar{P}_{bh}), \tag{1}$$

$$R_2 = \bar{q}_G - J_G\,(P_{res} - \bar{P}_{bh}) - \bar{q}_{gl}, \tag{2}$$

$$R_3 = \bar{P}_{an} - \kappa\,P_{gas}, \tag{3}$$

$$R_4 = \bar{q}_{gl} - 0.0075\,C_V\sqrt{\rho_G(\bar{P})\,\langle \bar{P}_{an} - \bar{P}_{inj}\rangle_+}, \tag{4}$$

where R_1 and R_2 are the liquid and gas balances between the linear inflow relation at the mean bottom-hole pressure and the mean rates at the top, R_3 is the static gas column from the platform to the base of the annulus ($\kappa = 1.138$ for this fluid and depth), and R_4 is the valve law written with the mean pressures, the only approximate relation of the four (the mean of a square root is not the square root of the mean, but the modulation of the valve pressure drop is within 25 % and the error is a few per cent), with a smooth gate that closes the valve when the mean annulus pressure falls below the mean injection pressure. The loss is the data term plus λ times the mean square of the four residuals, each scaled by the standard deviation of the quantity it compares, evaluated on 200 Latin-hypercube points (McKay, Beckman and Conover, 1979) of the plane of the decisions; $\lambda = 1$, and values from 0.1 to 100 give the same fit with residuals decreasing with λ. The residuals are closed-form functions of the network outputs, so that the gradient of the physics term is exact and cheap, unlike the detached-Jacobian construction that the compositional surrogates needed (Carvalho et al., 2026). The information in the physics is therefore not the dynamics but the accounting: the network is told that whatever enters the well on average leaves it on average, and that the lift gas is set by two mean pressures it also predicts.

5. Results

Table 2 and Fig. 3 compare the two networks on the 45 test runs. The data-only network predicts the mean gas rate at the top and the mean gas-lift rate with R^2 of 0.99 and the three mean pressures with 0.986 to 0.998, but the mean liquid rate only with 0.71: that rate varies little over the plane (3.5 to 5.3 kg s^{-1}) and its time average over three cycles of a surging signal is the noisiest of the targets. The physics-informed network raises it to 0.89, with the root-mean-square error falling from 0.29 to 0.18 kg s^{-1}, because the balance R_1 ties it to the mean bottom-hole pressure, which is smooth; the gas and lift-gas rates reach 0.997 and 0.998 and the scaled residuals fall by a factor of four to ten. The spans are the weak point of both networks: the liquid span, which carries the surge, reaches 0.73 and 0.87; the gas span and the span of the injected gas stay below 0.65, since the maximum of a surge in three cycles is a noisy marker by nature. Fig. 4 shows the physics-informed surrogate on the plane of the decisions: the mean liquid rate rises with the lift gas above the boundary and falls as the choke closes, the lift-gas rate is zero below 90 bar and rises with P_{gas} as the valve law dictates, and the span of the liquid rate marks the slugging region with a narrow band of artefacts along the boundary, where the data change class within a few bar.

With few data the physics counts more. Table 3 and Fig. 5 retrain both networks on 10, 20 and 40 runs drawn from the training set, three draws each, and test them on the same 45 runs. The physics-informed network has the lower error for the gas and lift-gas rates at every size — about half the error of the data-only network with 10 and 20 runs — and for the liquid rate from 20 runs on; with 40 runs it reaches the liquid error that the data-only network does not reach with 105. The hyperparameter grid confirms the picture: of 24 architectures, one data-only network and nine physics-informed networks reach a mean R^2 of 0.8 over the six requested outputs, and the best physics-informed networks are small (two layers of 8 or 16 units).

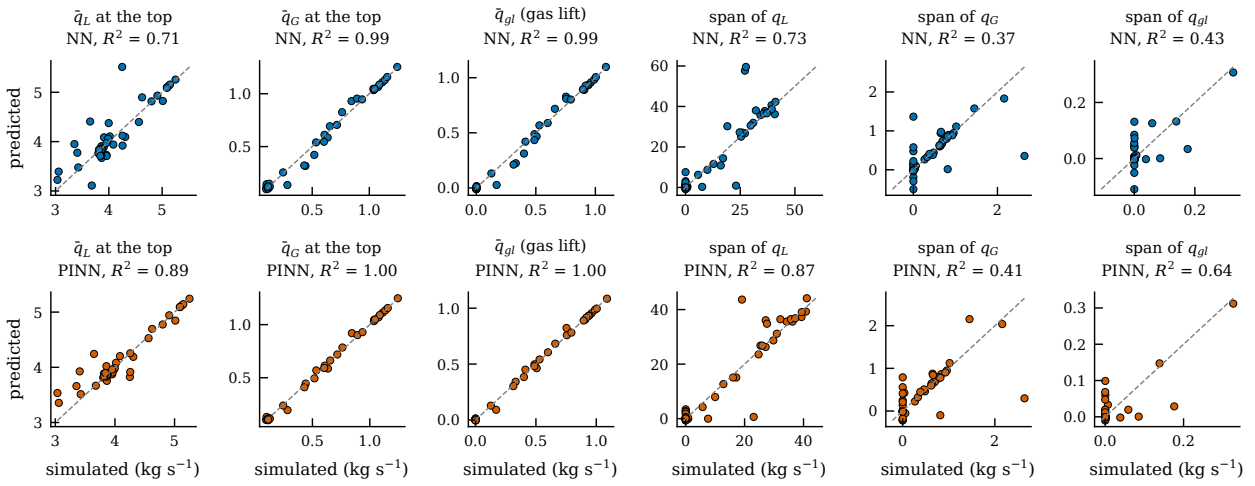


Figure 3: Test parity of the six requested outputs for the data-only network (top row) and the physics-informed network (bottom row).

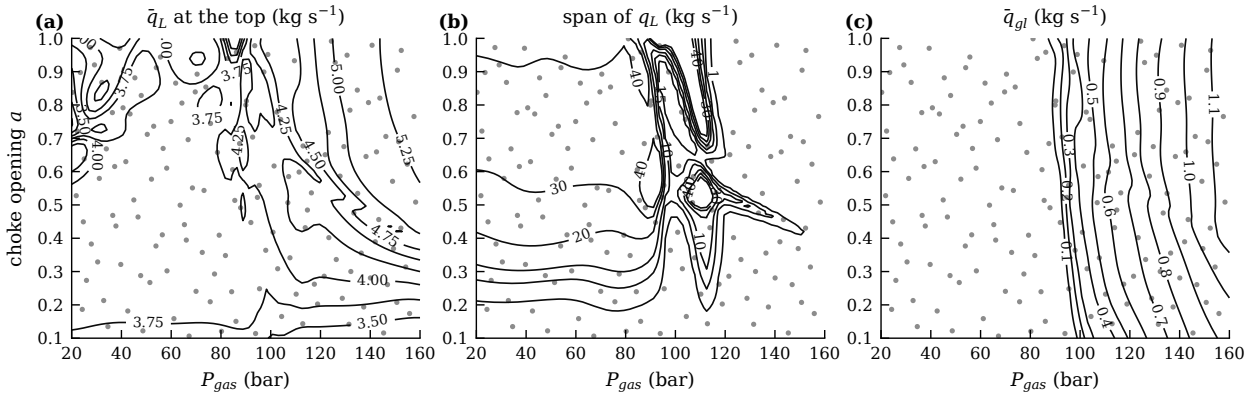


Figure 4: The physics-informed surrogate on the plane of the decisions as labelled level curves, with the positions of the simulated runs (grey dots): (a) mean liquid rate at the top; (b) span of the liquid rate, the signature of slugging; (c) mean gas-lift rate.

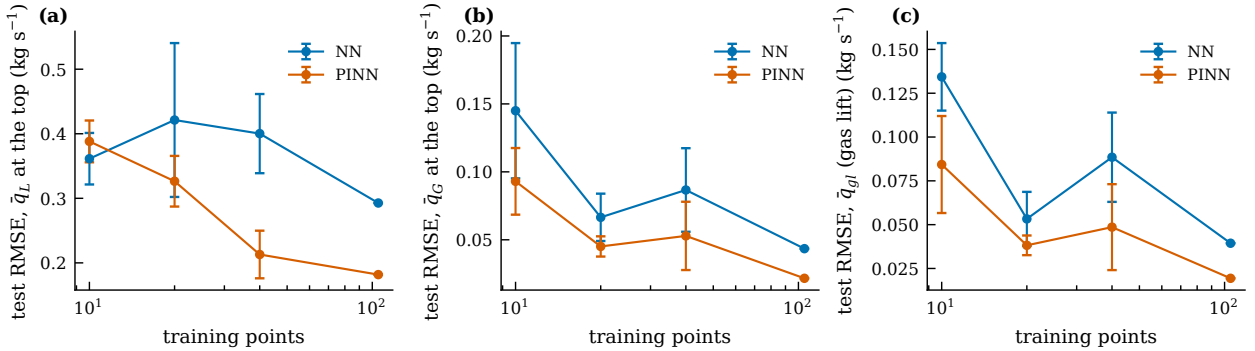


Figure 5: Learning curves of the two surrogates: test root-mean-square error of (a) the mean liquid rate, (b) the mean gas rate and (c) the mean gas-lift rate against the number of training runs (mean and standard deviation over three draws; the last point is the full training set).

Table 2
Test metrics of the two surrogates.

output	NN R^2	NN RMSE	PINN R^2	PINN RMSE
$\bar{q}_L$ at the top	0.708	0.293	0.887	0.182
$\bar{q}_G$ at the top	0.988	0.043	0.997	0.022
$\bar{q}_{gl}$ (gas lift)	0.990	0.039	0.998	0.019
span of q_L	0.733	7.907	0.869	5.543
span of q_G	0.373	0.450	0.415	0.435
span of q_{gl}	0.434	0.044	0.645	0.035
$\bar{P}_{bh}$	0.986	1.397	0.974	1.950
$\bar{P}_{inj}$	0.986	1.395	0.983	1.549
$\bar{P}_{an}$	0.998	1.650	0.999	0.953
residuals R_1–R_4 (RMS, std units)	–	0.615, 0.020, 0.033, 0.054	–	0.135, 0.017, 0.021, 0.031

Test set of 45 runs (105 for training); RMSE in kg/s for the flow rates and bar for the pressures. Residuals: root-mean-square of the four scaled physics residuals on 400 held-out collocation points. Both networks have two hidden layers of 32 hyperbolic-tangent units; the PINN adds the physics term with $\lambda = 1$ on 200 Latin-hypercube collocation points.

Table 3
Sparse-data study: test error of the mean rates against the number of training runs.

training points	$\bar{q}_L$ NN	$\bar{q}_L$ PINN	$\bar{q}_G$ NN	$\bar{q}_G$ PINN	$\bar{q}_{gl}$ NN	$\bar{q}_{gl}$ PINN
10	0.361 ± 0.040	0.388 ± 0.032	0.145 ± 0.050	0.093 ± 0.025	0.134 ± 0.019	0.084 ± 0.028
20	0.421 ± 0.119	0.326 ± 0.039	0.067 ± 0.017	0.045 ± 0.007	0.053 ± 0.015	0.038 ± 0.006
40	0.400 ± 0.061	0.213 ± 0.037	0.087 ± 0.031	0.053 ± 0.025	0.088 ± 0.025	0.049 ± 0.025
105	0.293	0.182	0.043	0.022	0.039	0.019

Test RMSE (kg/s), mean ± standard deviation over three random subsets of the training set; last row: the full training set.

6. Operational optimisation

[Section on hold: the formulation below is set; the optimisation runs and their figures are pending.] The optimisation takes three things from the preceding sections. First, the physics-informed surrogate of Section 4.2, which returns for any (a, P_{gas}) the mean liquid, gas and lift-gas rates and their spans in a fraction of a millisecond, with the balances of the well and of the branch built in; its weights are stored with the campaign and the optimiser calls it directly. Second, the stability map of Section 3.4, which bounds where the surrogate is trusted: the boundary between the cycling and the steady regions is where the markers change fastest and where the spans carry the surges, and the six budget-limited runs mark the strip where the simulator itself is least reliable. Third, the formulation. The operator maximises the revenue of the mean liquid rate minus the cost of compressing the lift gas from the separator pressure to P_{gas} — the mean gas-lift rate and P_{gas} set the compressor power — subject to the capacity of the separator for liquid surges, expressed on the span or the peak of the liquid rate, and to a bound on the pressure swing at the wellhead; both constraints have operational meaning and replace an arbitrary penalty on slugging. Over the plane of the two decisions this is a two-variable problem that can be solved by enumeration on the surrogate and verified by the simulator at the optimum, and the comparison with the optimum of a steady-state model at the same points will measure the bias of nodal optimisation on an unstable well. The one open point that the optimisation inherits is the quality of the spans: the surrogate ranks the surges but does not predict them to better than a quarter, so the surge constraint will be formulated on the class and on the liquid span, the two markers the surrogate resolves, and confirmed by simulation where it is active.

7. Conclusions

A dynamic two-fluid simulation of a gas-lifted well can feed a surrogate of its operating decisions once the campaign is designed for it: warm-started chains along the shortest path through the sampled points, an adaptive horizon that recognises the established regime and captures three cycles, and markers at characteristic points instead of trajectories. Over the plane of the choke opening and the platform gas pressure the well has a stability boundary from 80 to 125 bar of lift gas, continuous injection modulated by the cycle below it and up to 34 % more production above it. Of the two surrogates trained on the 156 runs, the physics-informed one, whose physics are the time-averaged mass balances of

the well and of the gas-lift branch, predicts the mean liquid rate with R^2 0.89 against 0.71 for the data-only network and the gas and lift-gas rates with 0.997, needs half the data for the same error, and leaves the spans of the rates as the quantity to be handled by the class and the stability map rather than by regression. The operational optimisation of the two decisions on this surrogate, under separator and pressure-swing constraints, is pending.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used AI-assisted tools for language editing and code assistance. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Data availability

The simulator used in this study is not publicly available; the campaign design (sampling, horizon, markers) and the surrogates are described in full so that they can be reproduced with an independent implementation, and the table of markers and the trained networks are available from the corresponding author upon reasonable request.

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